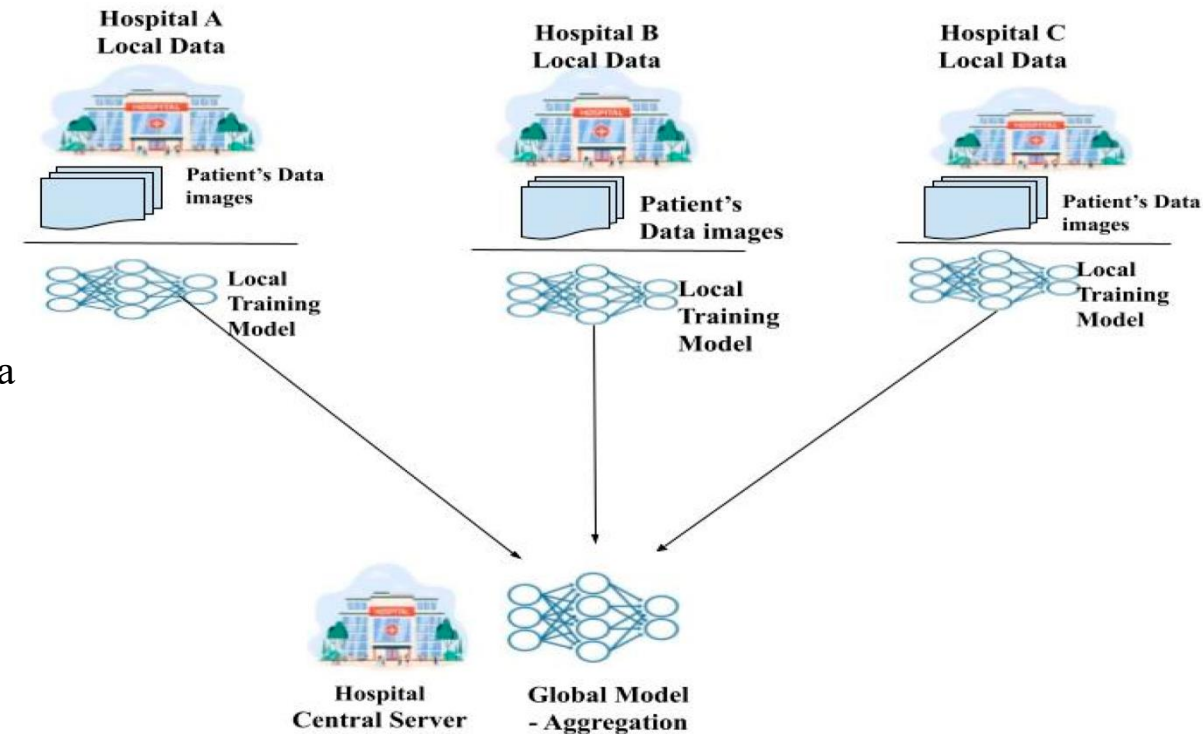


Federated Learning in Bioinformatics: An Overview

Francesco Gualdi , PostDoc researcher, IDSIA - USI/SUPSI

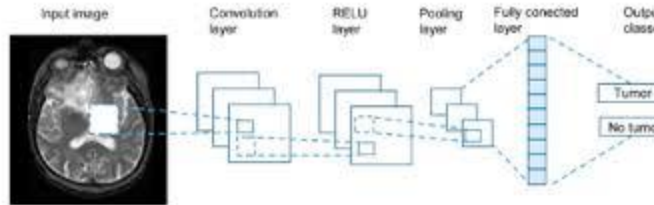
Why FL?

- Centralizing clinical records for deep learning raises concerns:
 - Privacy risks
 - Data ownership issues
 - Legal restrictions
- **Balancing the need for shared knowledge with protecting sensitive medical data.**
- **Privacy-preserving techniques** offer a solution to use multi-center data securely.
- **Federated Learning (FL):**
 - Trains large ML models across multiple centers
 - No need to share raw patient data
 - Preserves privacy and security

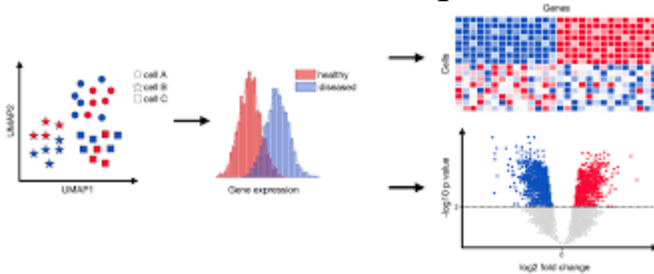


Current applications in Bioinformatics

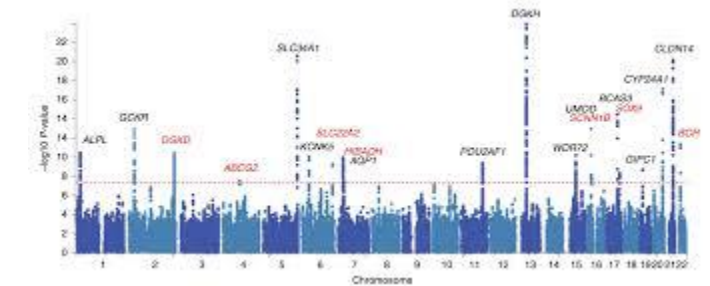
Medical Imaging



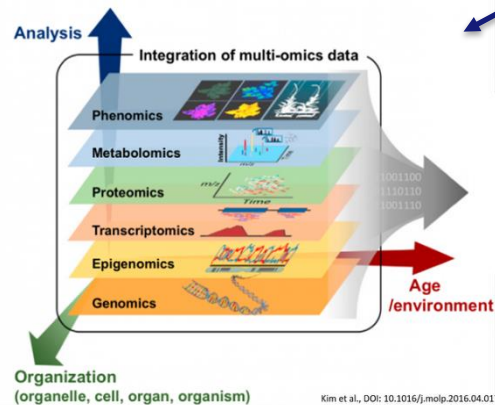
RNA-seq



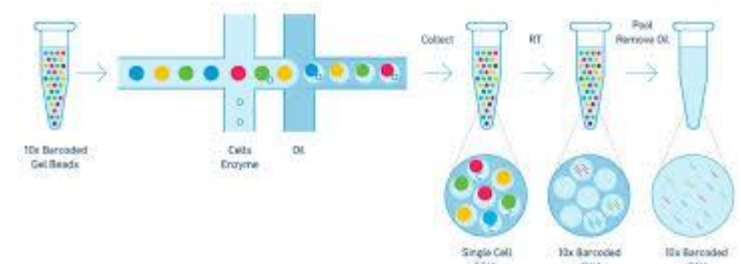
Genome Wide association Studies (GWAS)



Multi omics



scRNA-seq

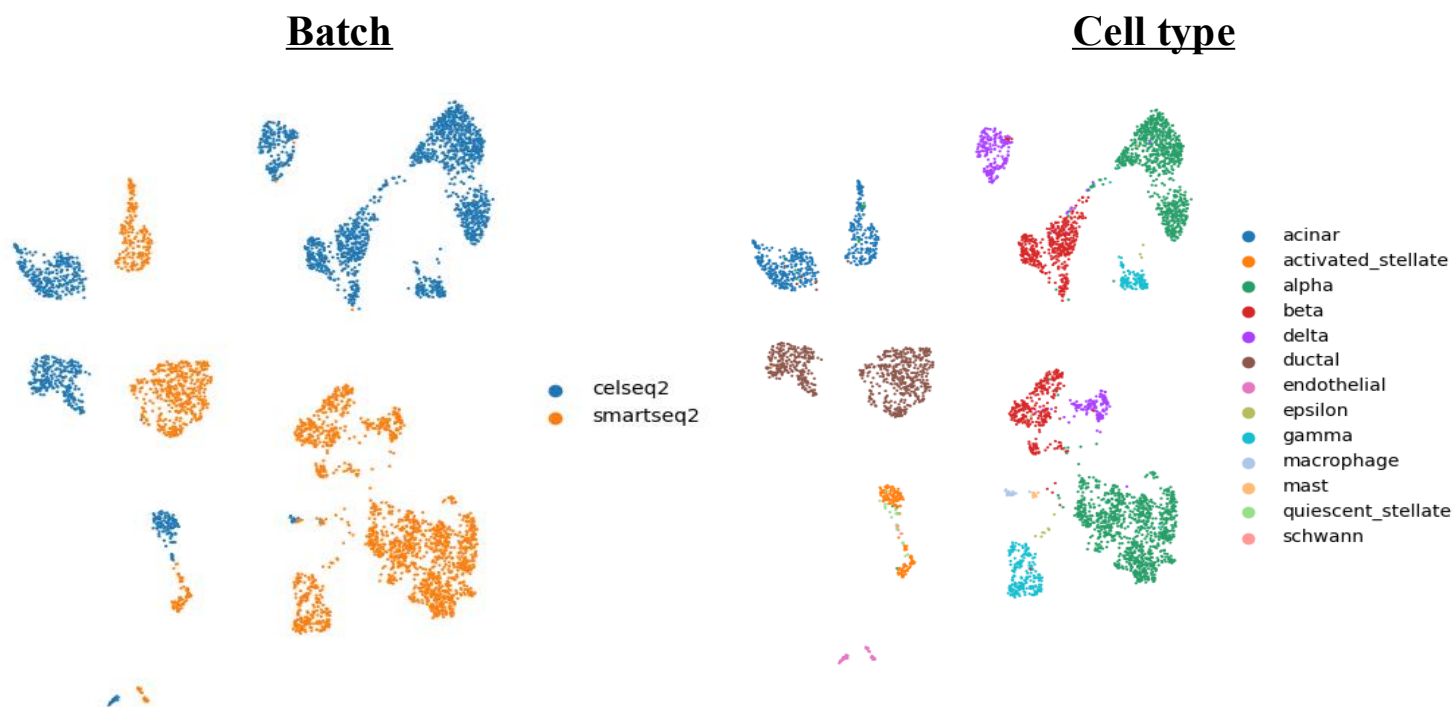


Key Challenges in FL applied to Bioinformatics

- Data heterogeneity (different labs, protocols, platforms) -> Batch effect
- Scalability (large cohorts, omics data) -> Computational challenges
- Privacy and Data generation costs -> Difficulty in data managing

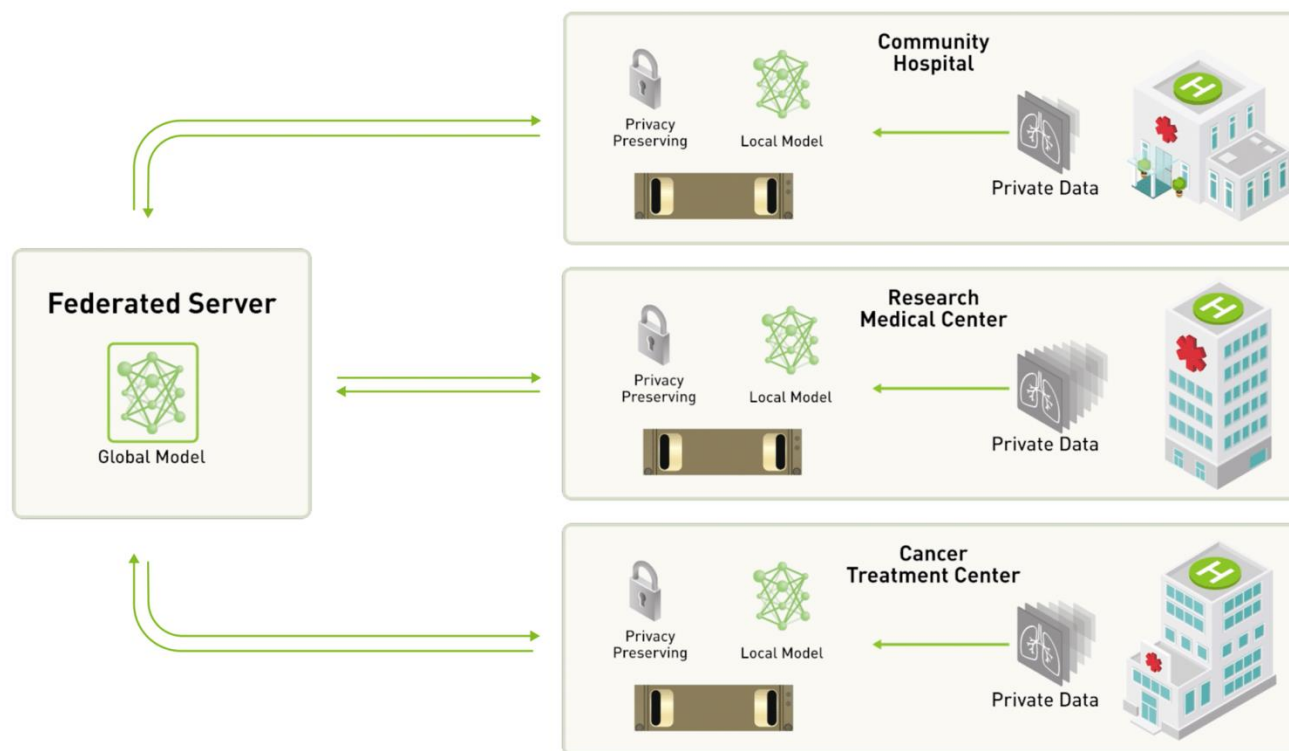
Data Heterogeneity

UMAP on scRNA-seq data



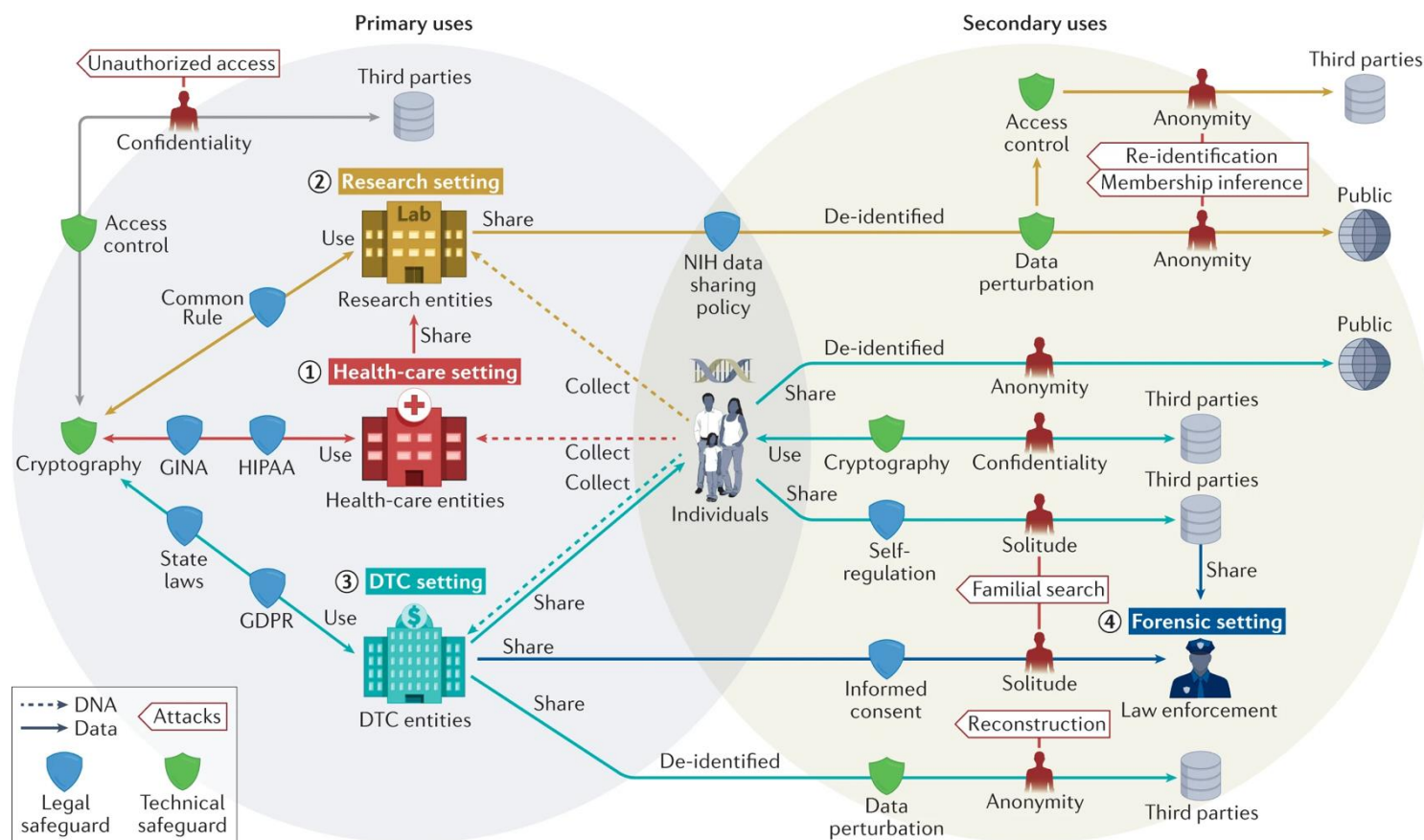
- Same Cell populations in different Clients have different distributions
- This can lead to biased results
- Need to batch mitigation strategies

Scalability



- **Communication overhead:** Exchanging model updates across many clients increases bandwidth and latency costs.
- **System heterogeneity:** Clients differ in computational power, storage, and network reliability.
- **Server bottleneck:** A central aggregator may struggle to handle massive simultaneous client connections

Privacy and data generation



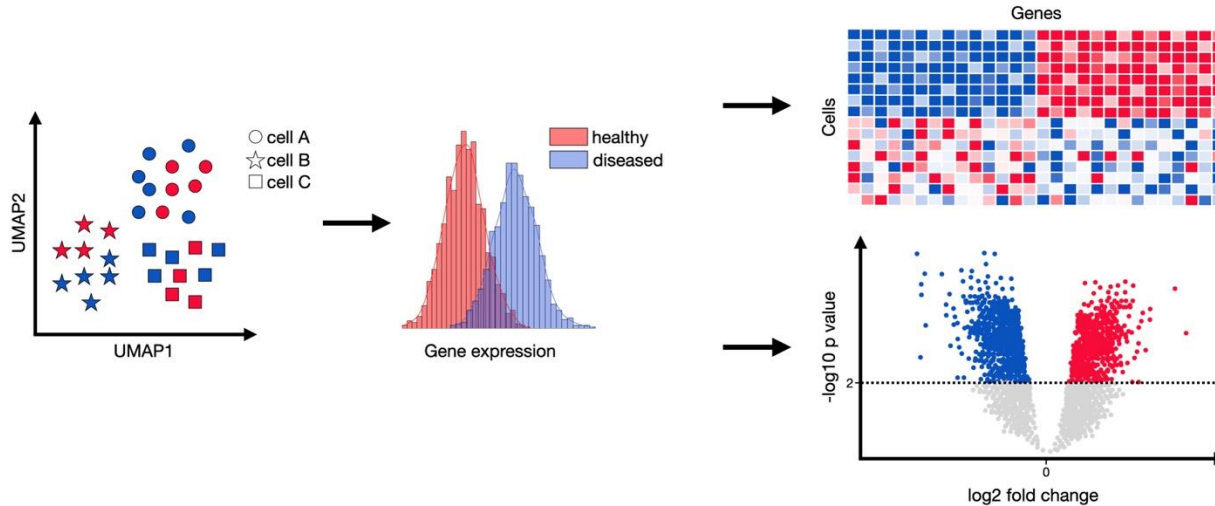
- Even with FL, model updates may leak private patterns (e.g., membership inference attacks, i.e. training an AI model in order to share sensible informations).
- Data generation has costs institutions are not keen in sharing it
- Companies using research data for profit in ways not aligned with participants' consent or public interest.

Applications of FL in Bioinformatics

Field	Main Task	Example FL Methods
Proteomics & Gene Expression	Differential expression, cell type classification	FedProt, HyFed (limma voom), DL (Flower, TFF)
GWAS	Association testing, scalable regression	FedGLMM, FedGMMAT, REGENIE (MPC/HE)
Single-cell RNA-seq	Cell type classification	scFed (ACTINN, SVM, XGBoost, GeneFormer)
Multi-omics	Diagnostics	Vertical FL, adaptive neural networks
Medical Imaging	Classification, segmentation	Federated labelling, harmonised feature learning

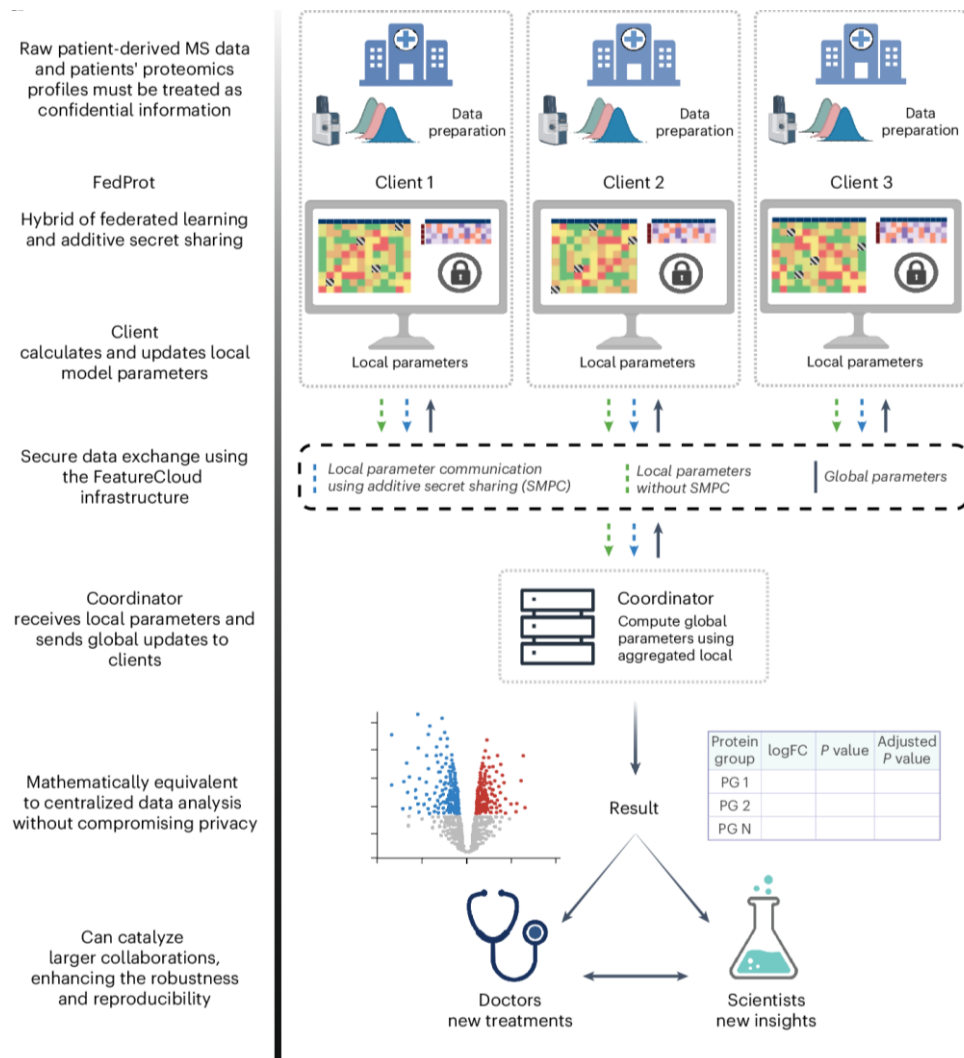
<https://doi.org/10.48550/arXiv.2503.09649>

Differential Expression in Proteins and Genes



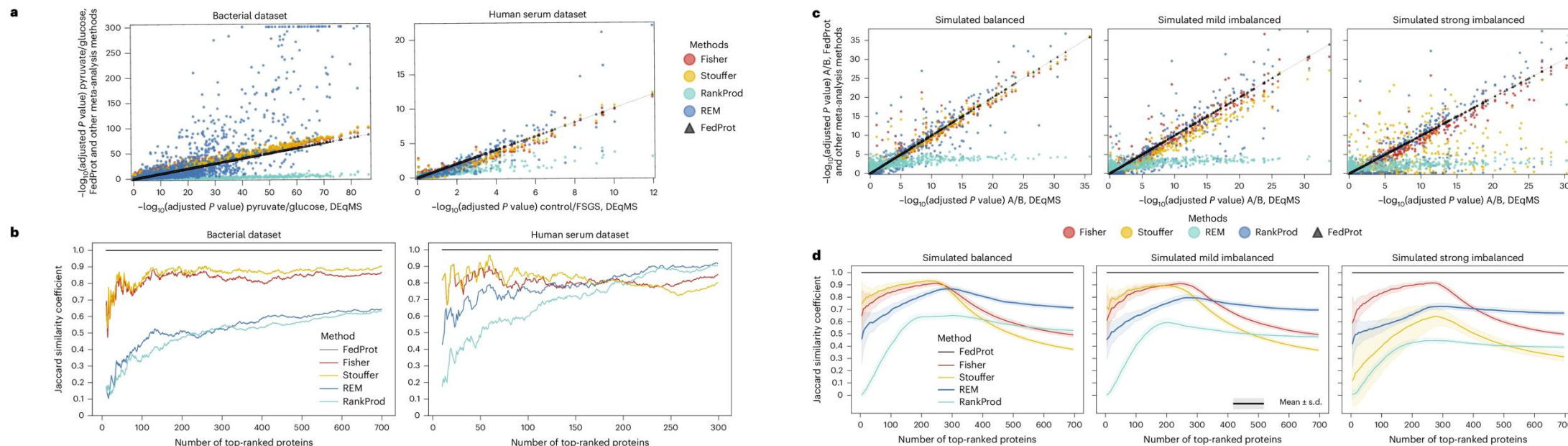
- Analysis of genes to identify those with significantly different expression levels between conditions (e.g., healthy vs. diseased tissue).
- Discover biomarkers, understand biological processes, and reveal molecular mechanisms of disease.

FedProt



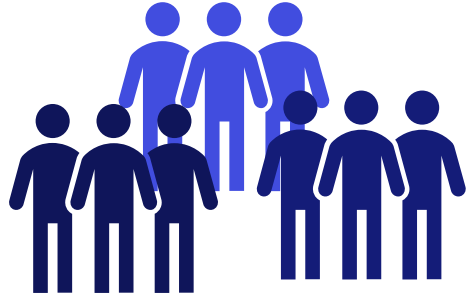
- Clients preprocess MS data; raw proteomics data stay private.
- Clients train a global model via parameter exchange, without sharing raw data.
- During client communication, privacy is maintained through secure multiparty computation
- Final output matches centralized analysis, with fold changes, confidence intervals, and adjusted p-values.

FedProt

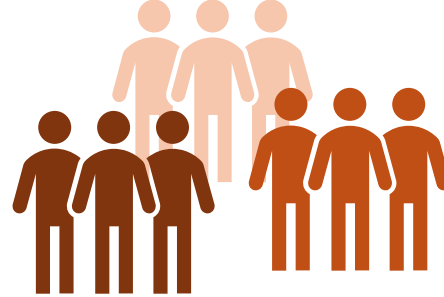


- **Comparison of P values:** $-\log_{10}(\text{BH-adjusted } P)$ between FedProt and other methods (y axes) and centralized DEqMS
- **Jaccard similarity:** Agreement in top-ranked proteins between centralized and decentralized approaches.

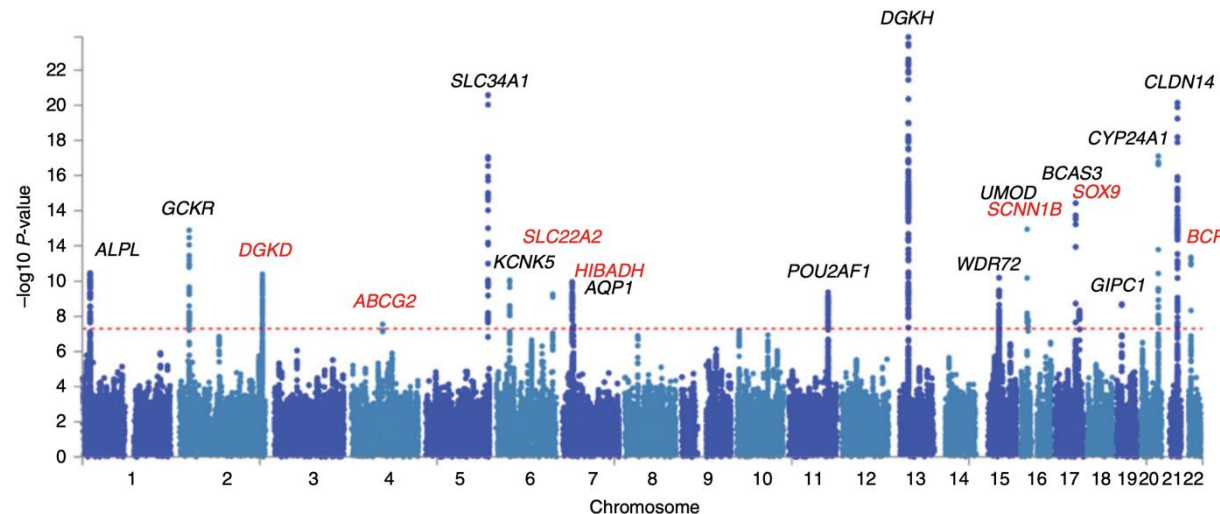
Genome Wide Association Studies (GWAS)



Control Group



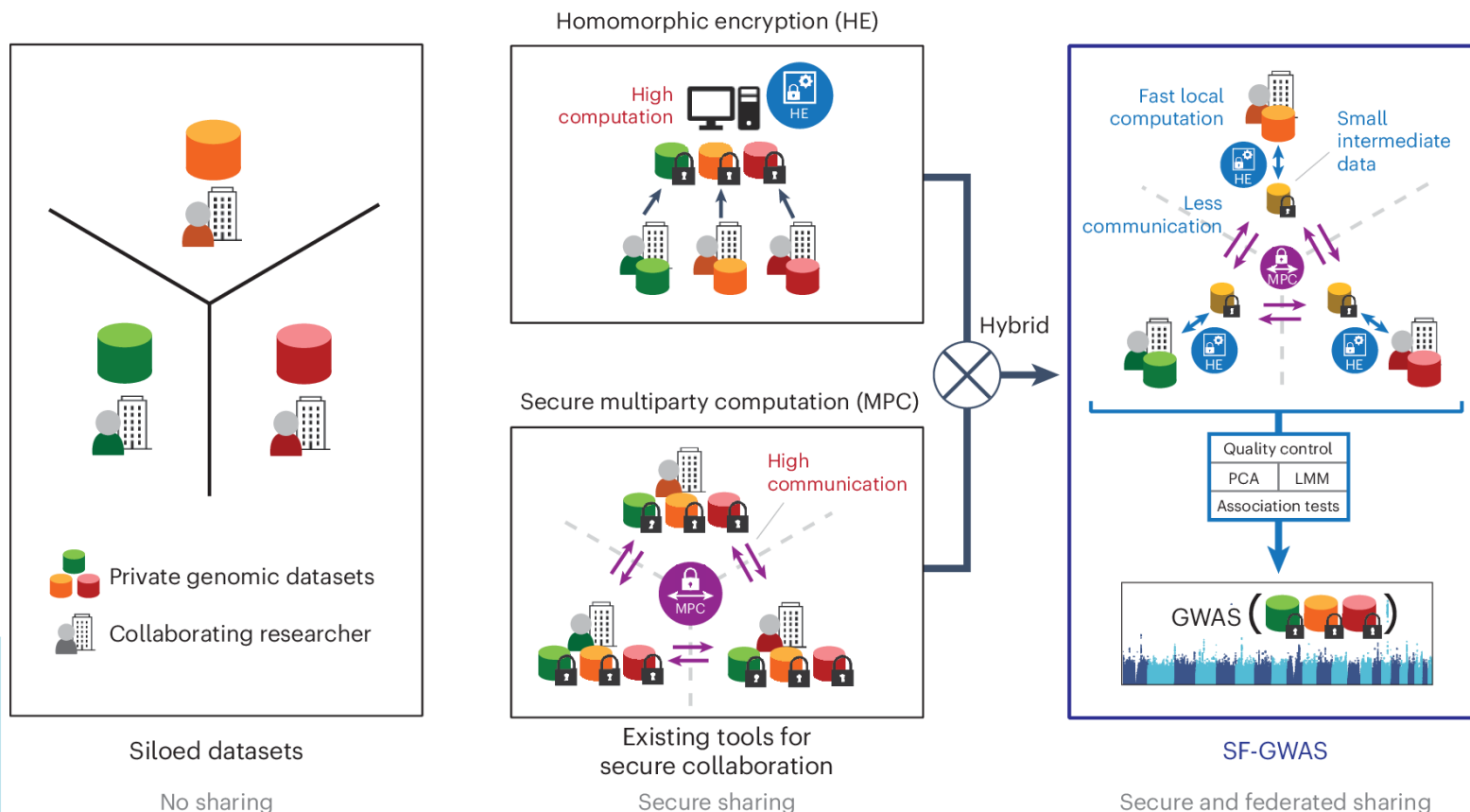
Case Group



- GWAS have successfully identified risk loci for a vast number of complex diseases
- Test association of every genotyped locus variant against a phenotype.
- The association is done by multiple regression using all subjects variants and phenotype.
- Regression allows to include other variables (covariates) on the equation.

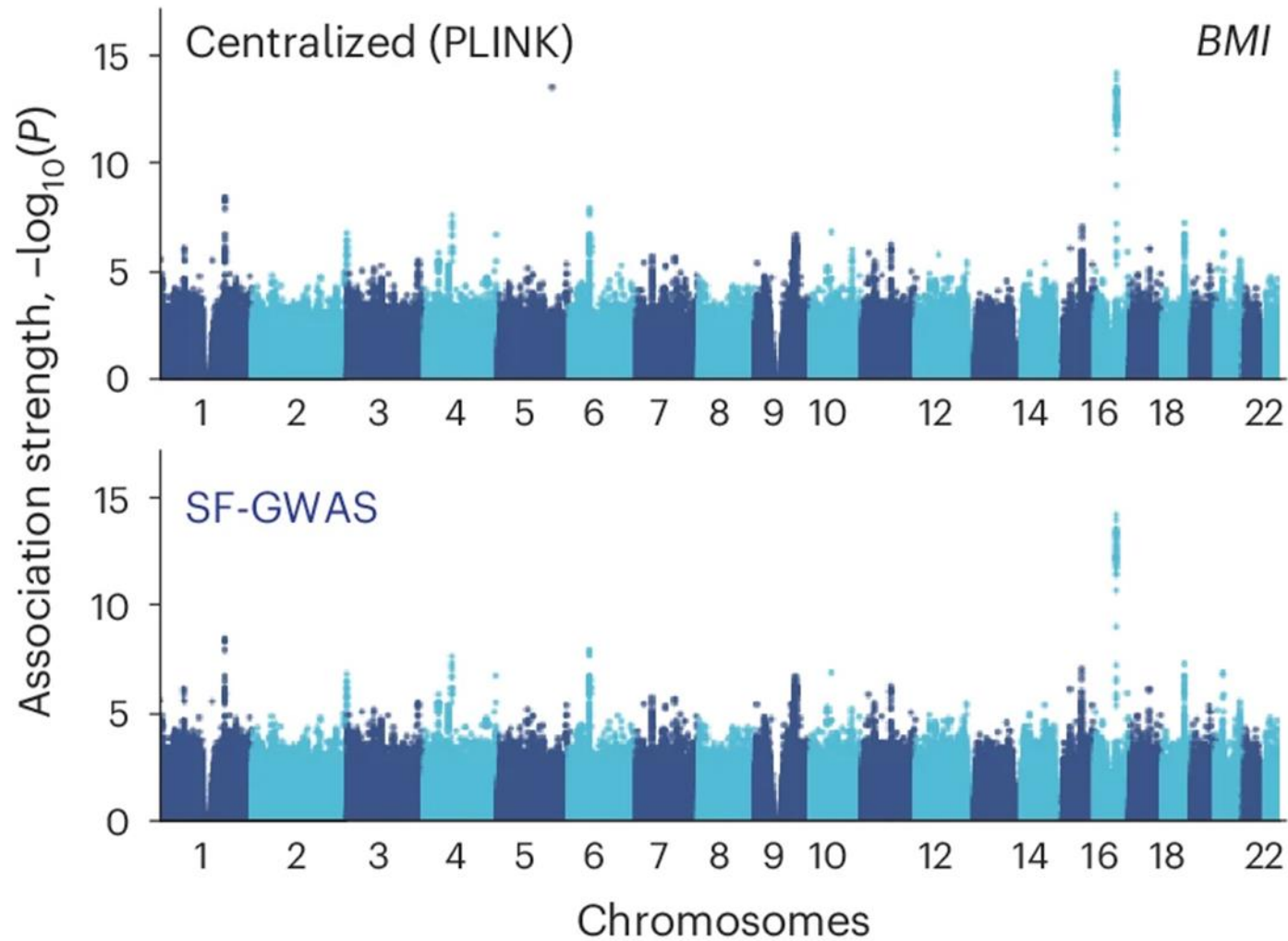
Howles, S.A., Wiberg, A., Goldsworthy, M. et al. Genetic variants of calcium and vitamin D metabolism in kidney stone disease. *Nat Commun* **10**, 5175 (2019). <https://doi.org/10.1038/s41467-019-13145-x>

Secure Federated (SF-GWAS)



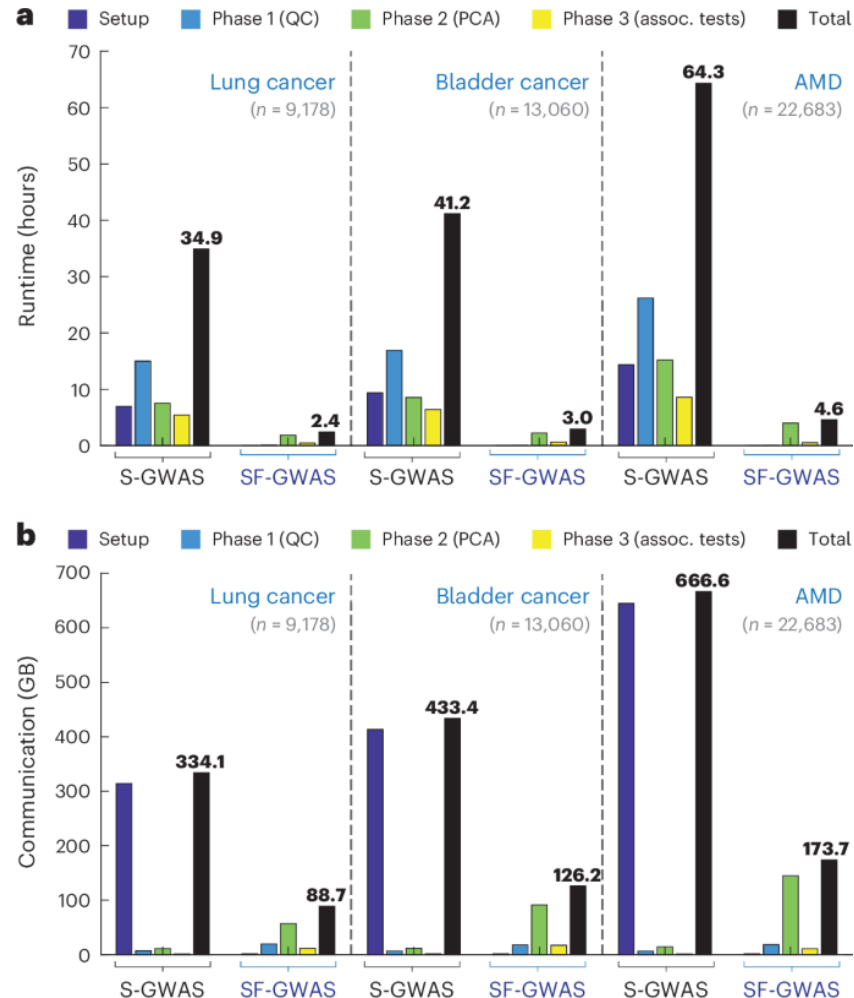
- SF-GWAS is a secure, federated algorithm for multisite genome-wide association studies.
- Enables collaborative genomic studies with cryptographic privacy guarantees.
- Each site keeps its genomic dataset locally, reducing data transfers and computational costs.
- Combines MPC (multiparty computation) and HE (homomorphic encryption) in a hybrid framework.
- improved precision and efficiency.

SF - GWAS



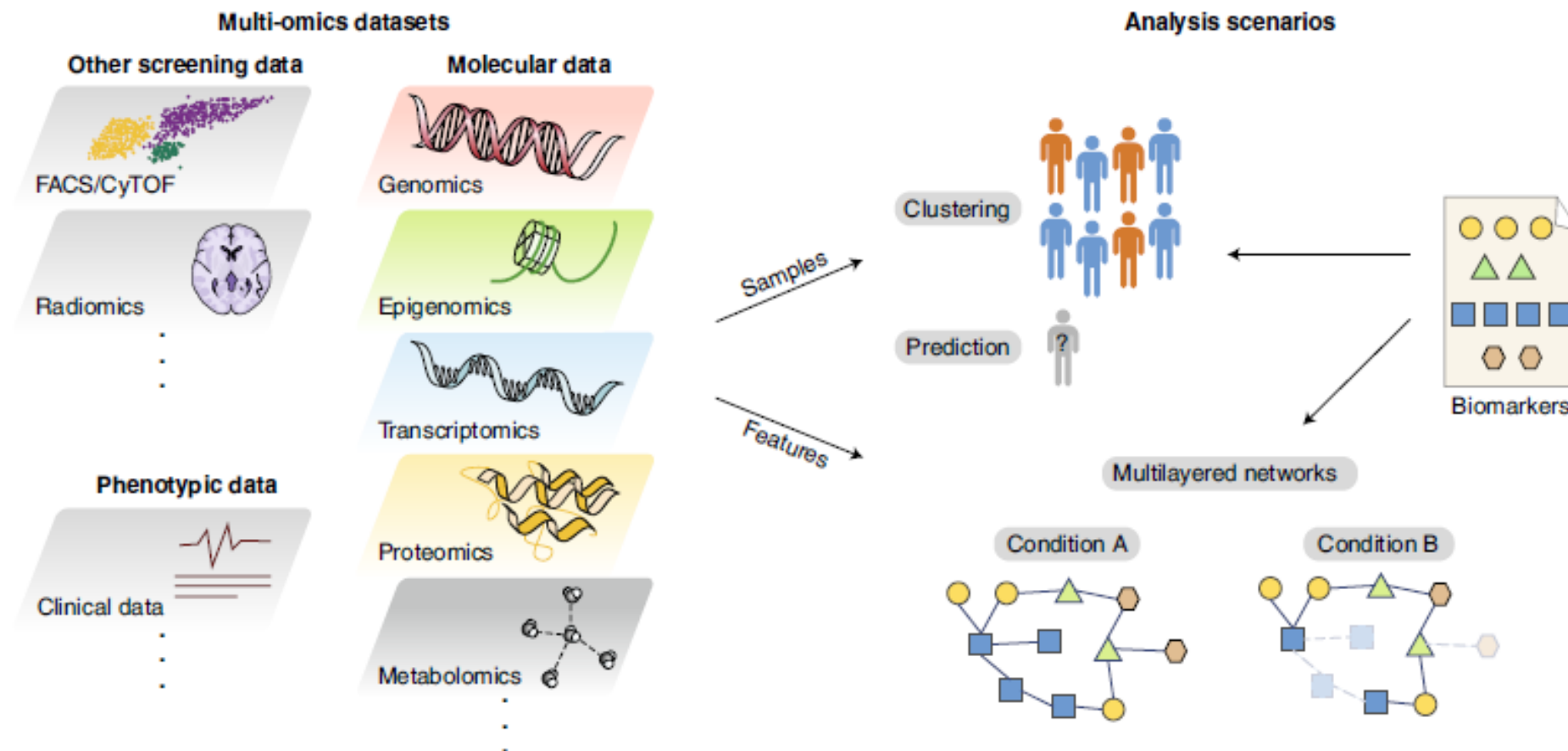
Replicates same results of centralized Dataset

SF - GWAS



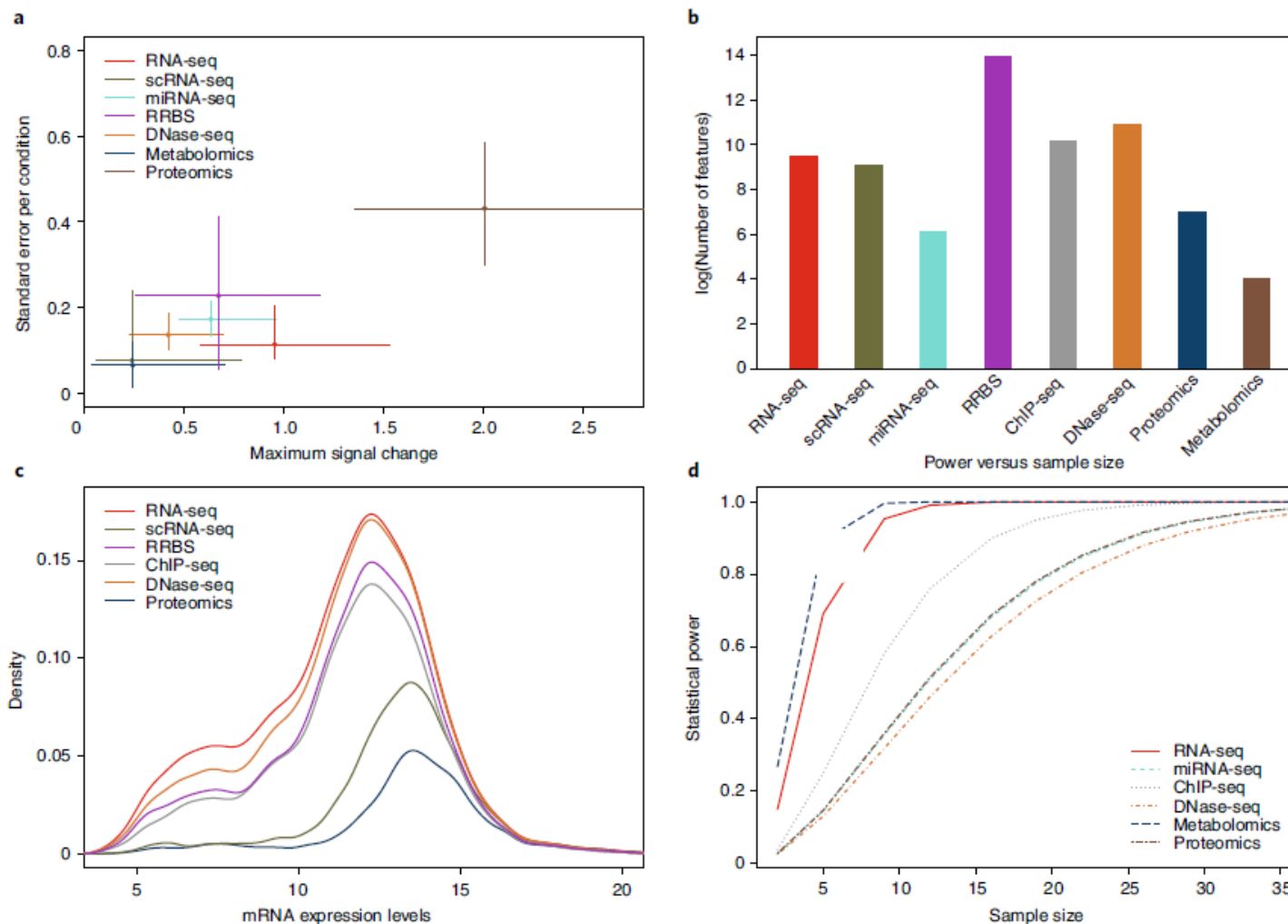
- Comparison with S-GWAS implementing cryptographic methods both implement full GWAS steps (QC, PCA, association tests).
- **Datasets:** Lung cancer, bladder cancer, and Age related Macular Degeneration; split across two machines to simulate collaboration.
- SF-GWAS avoids the encrypted data transfer required by S-GWAS, thanks to its federated design.
- **Results:** SF-GWAS is $\sim 10 \times$ faster and $\sim 3.5 \times$ more communication-efficient.

Multi - Omics



Tarazona, S., Arzalluz-Luque, A., & Conesa, A. (2021). Undisclosed, unmet and neglected challenges in multi-omics studies. *Nature Computational Science*, 1(6), 395–402. <https://doi.org/10.1038/s43588-021-00086-z>

Multi - Omics



Comparison of the properties of omics data types.

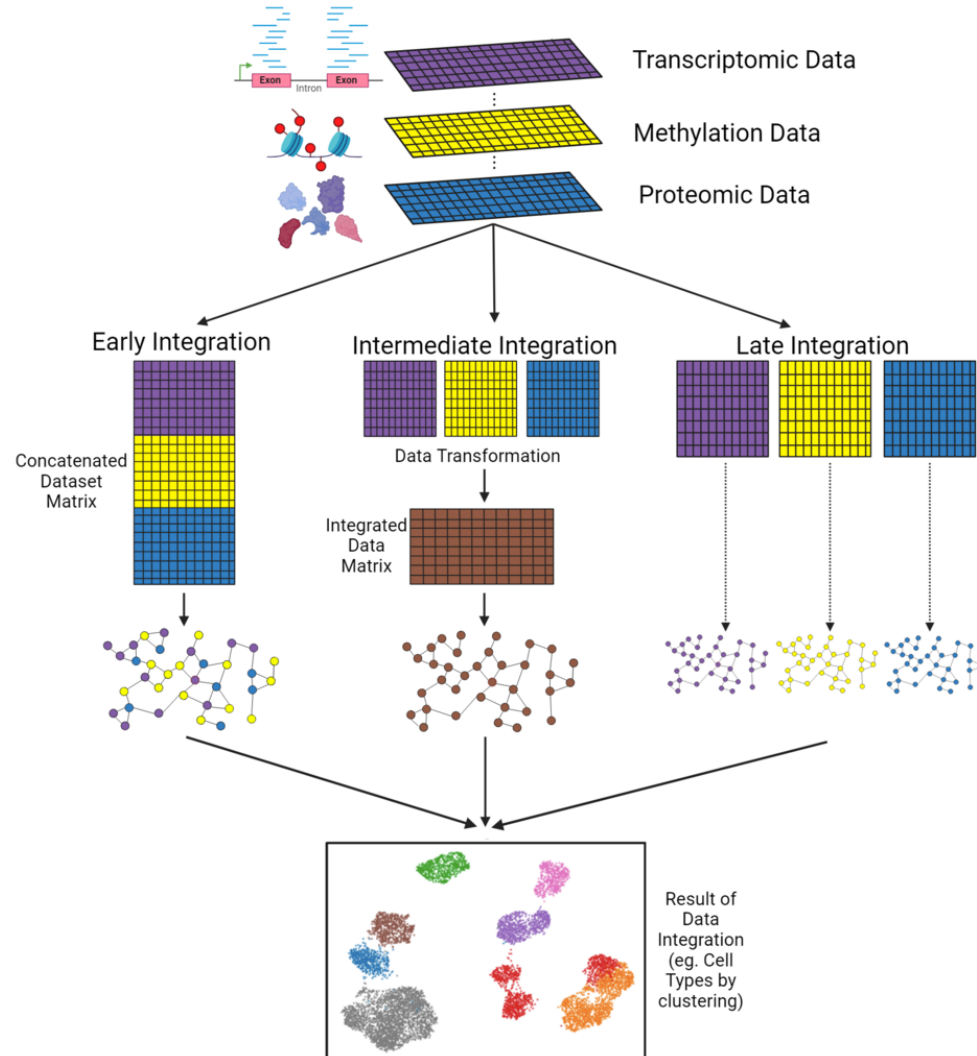
A: Signal-to-noise plot. Segments represent the interquartile ranges of the maximum signal change

B: Number of features detected by each technology.

C Differential coverage of the feature space. Each density line represents the distribution of the fraction of expressed genes captured by each method.

D: Statistical power curves across omics data types as a function of sample size

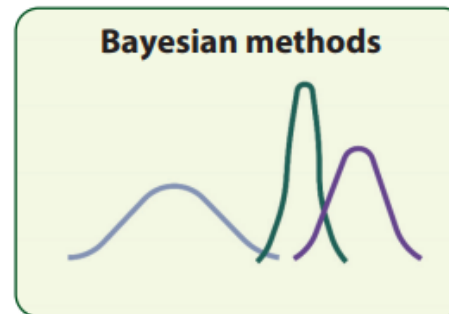
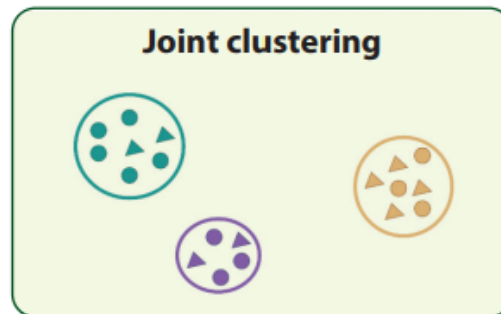
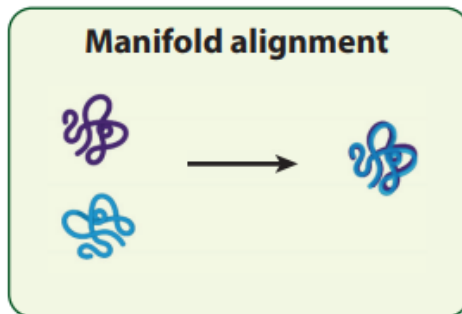
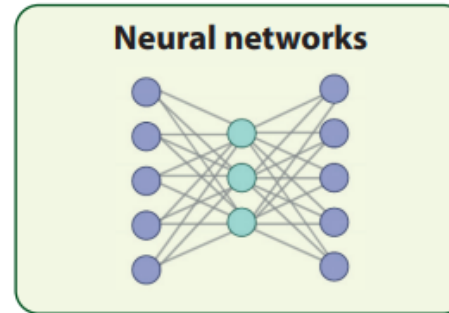
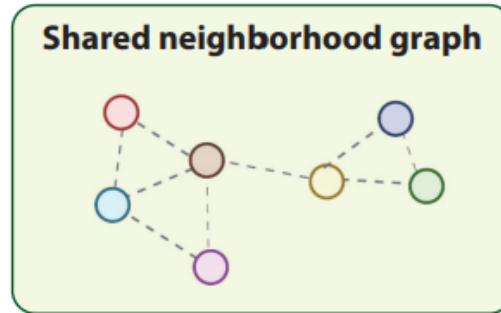
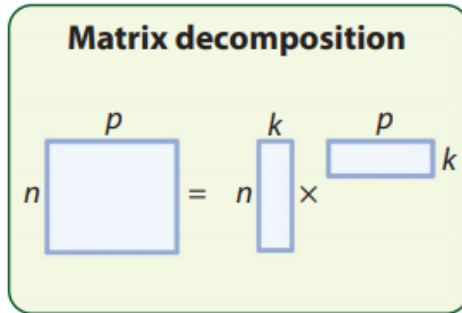
Multi – Omics integration strategies



- **Early integration (concatenation-based)**
 - Combine all features into one matrix.
 - Apply machine learning/statistical models.
 - Simple, but risks overfitting and ignores data-type differences.
- **Intermediate integration (transformation-based)**
 - Extract features (e.g., latent factors via PCA, NMF, autoencoders).
 - Align data into a shared space.
 - Keeps omics-specific structure while enabling comparison.
- **Late integration (model-based)**
 - Analyze each omics layer separately.
 - Combine results at decision or model level

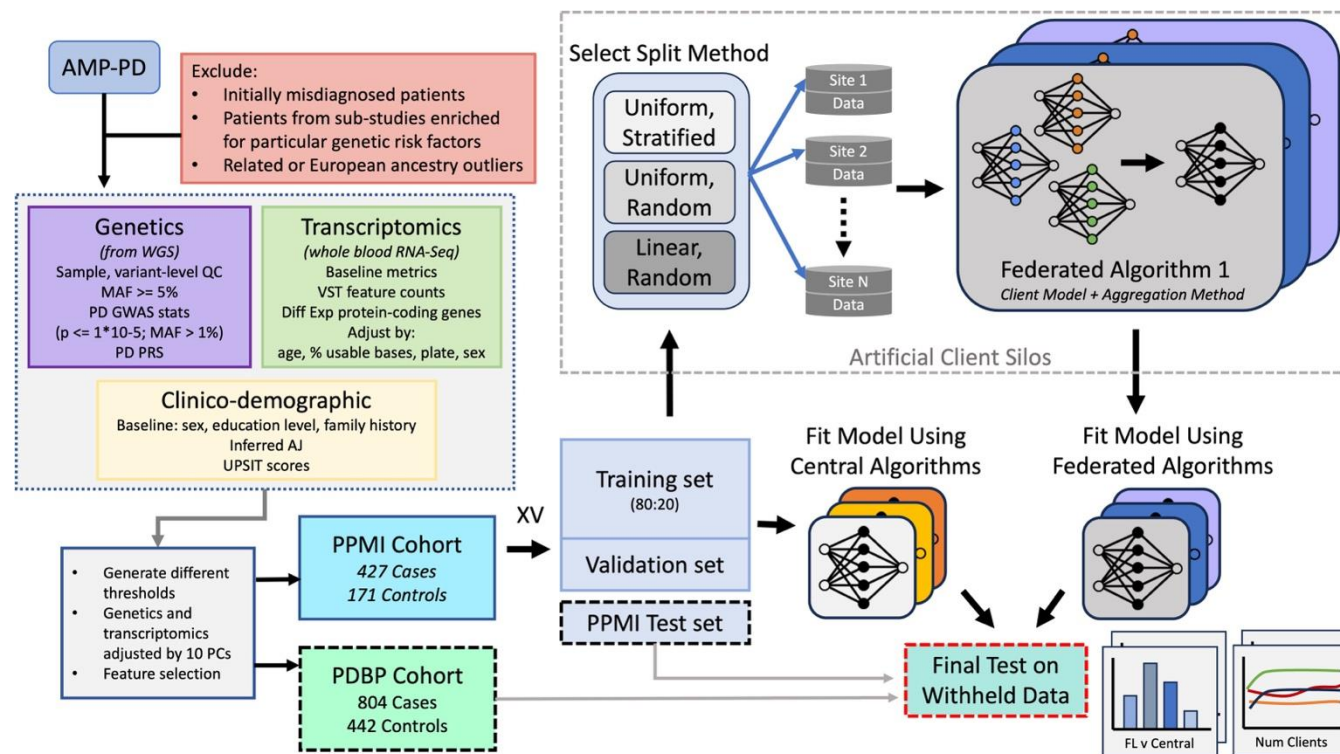
Flexible but may miss cross-omics interactions.

Multi – Omics – Integration strategies



- Active Research Field
- Many methods allow to multi-omics integration
- Applications:
 - Patient stratification in cancer and rare diseases
 - Biomarker discovery for diagnostics
 - Understanding molecular mechanisms
 - Precision medicine and drug response prediction

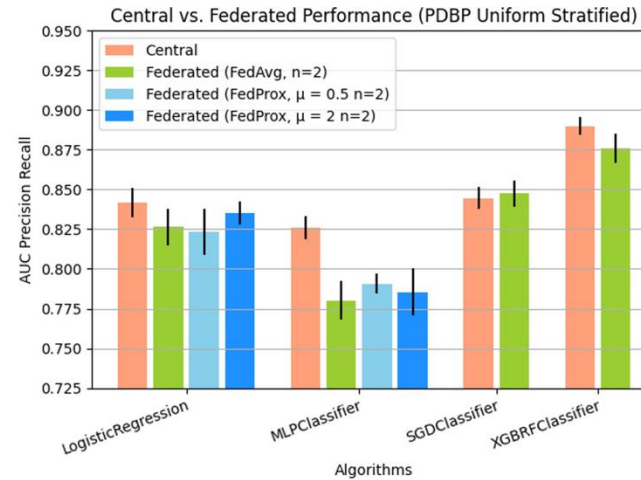
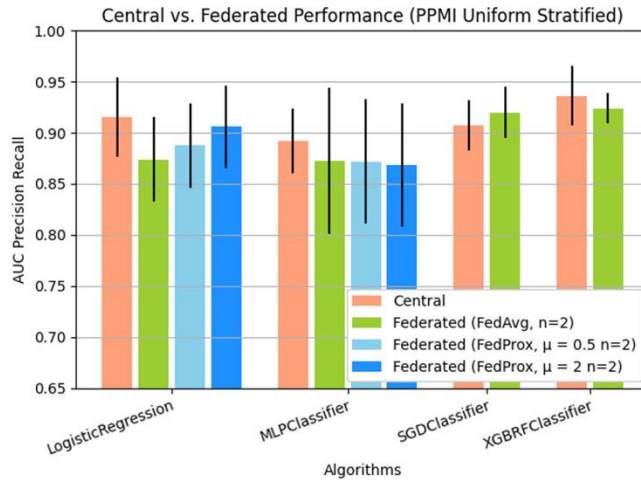
Multi – Omics FL



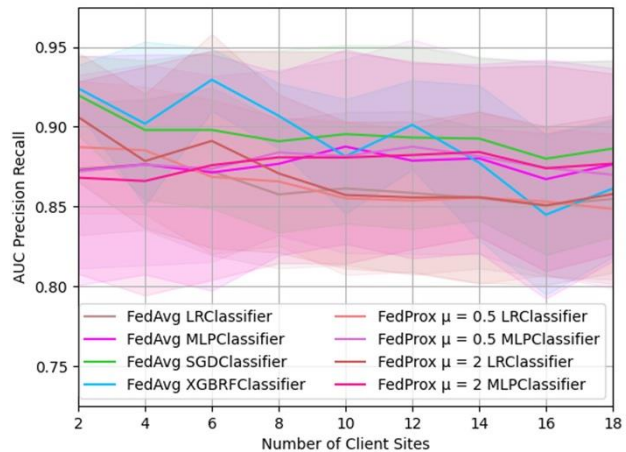
- Early integration of different features
- The study used Multi Omics Parkinson data to predict severity of the condition
- Comparison between Federated and Centralized Machine Learning models

Xu, J., Glucksberg, B. S., Su, C., Walker, P., Bian, J., & Wang, F. (2021). Federated Learning for Healthcare Informatics. *Journal of Healthcare Informatics Research*, 5(1). <https://doi.org/10.1007/s41666-020-00082-4>

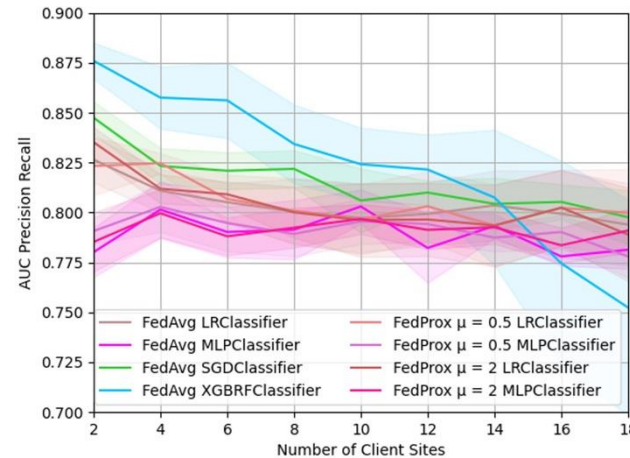
Multi – Omics FL



FL Algorithm Performance vs. Client Sites (PPMI Uniform Stratified)



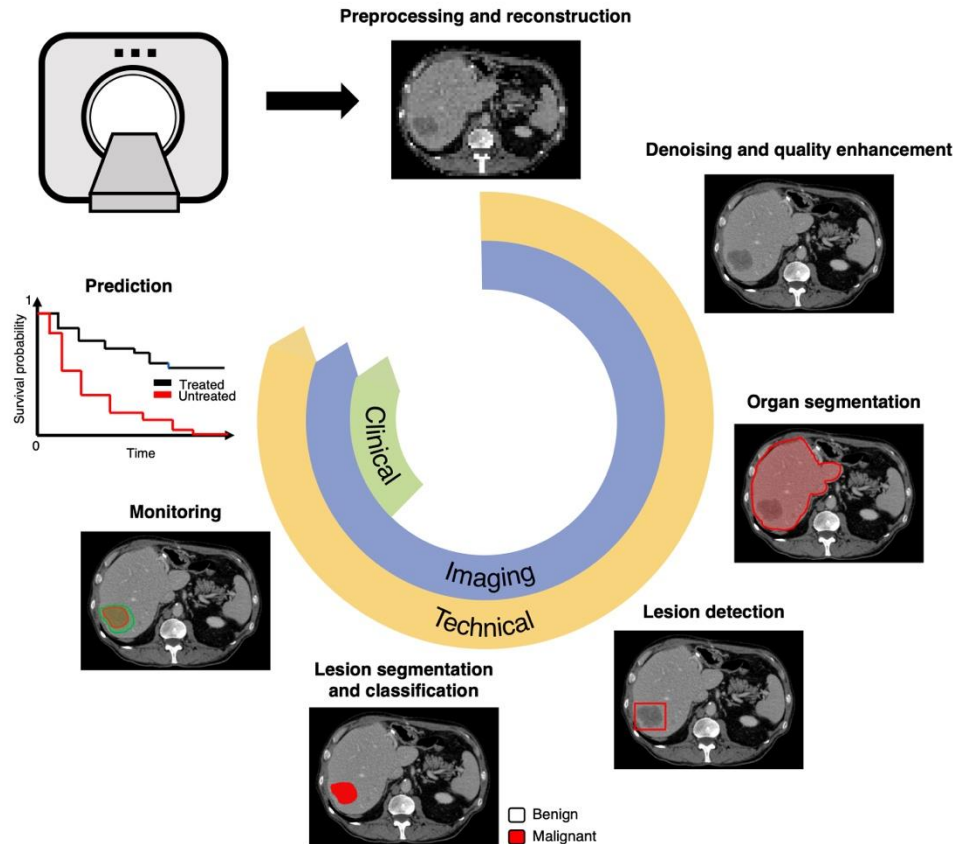
FL Algorithm Performance vs. Client Sites (PDBP Uniform Stratified)



- Federated and Centralized model have comparable performances

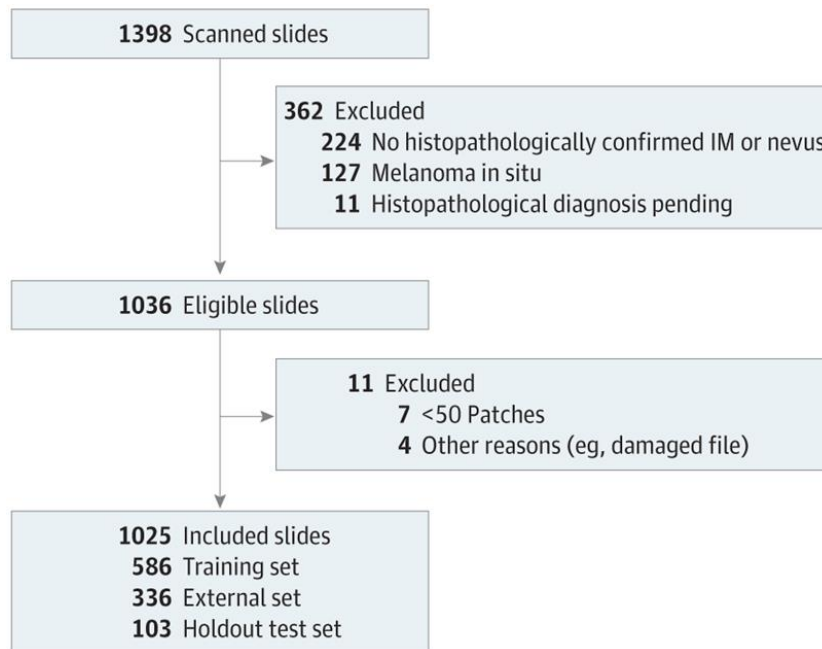
- Sample dispersion among client sites negatively impacts global model performance

Medical Imaging



- **Preprocessing & Reconstruction** – denoising, super-resolution, accelerating MRI/CT acquisition.
- **Segmentation** – automatic delineation of organs, tumors, or cells (e.g., U-Net in medical scans).
- **Classification** – labeling images or regions (healthy vs. diseased, cancer subtype).
- **Detection** – locating and identifying objects of interest (lesions, nodules, anomalies).
- **Feature Extraction & Radiomics** – quantifying shapes, textures, or biomarkers for predictive modeling.

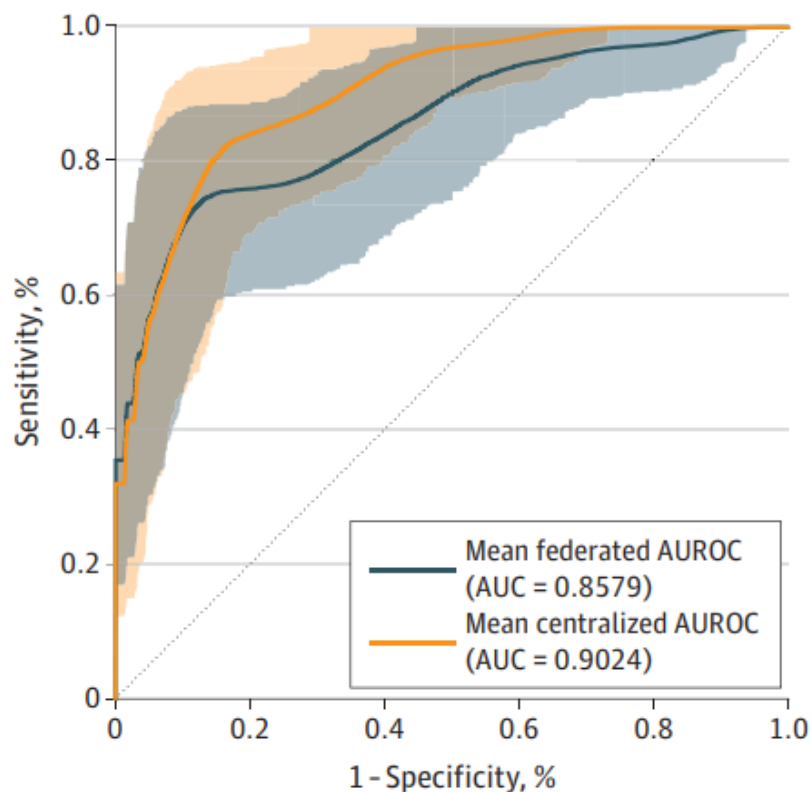
Medical Imaging - FL



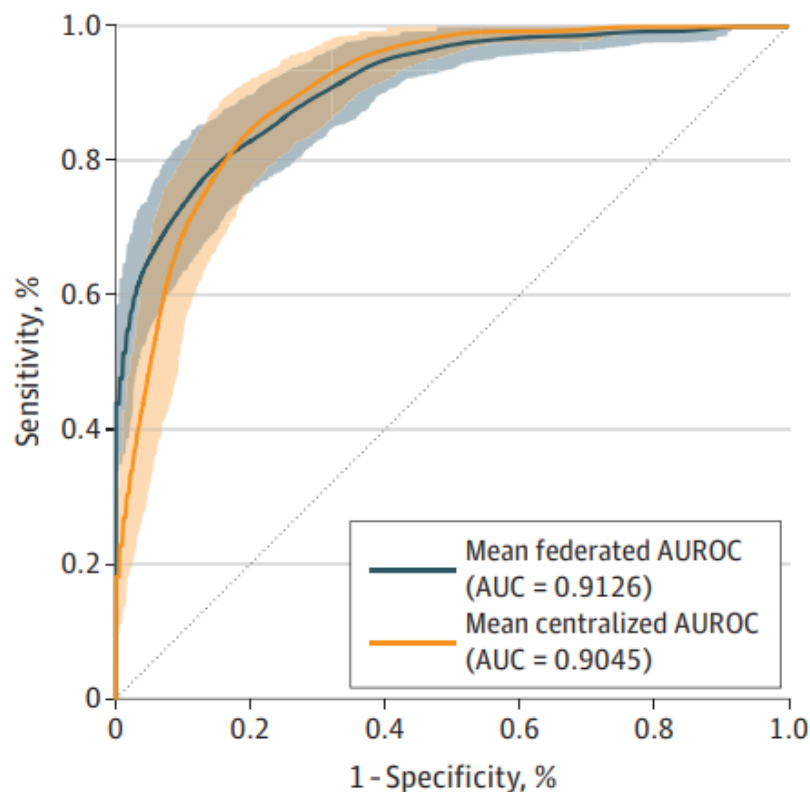
- FL for AI-based melanoma diagnostics.
- A federated model for melanoma-nevus classification was developed using histopathological whole-slide images from 6 German university hospitals
- FL can achieve comparable diagnostic performance to centralized learning while improving privacy protection

Medical Imaging - FL

A Federated vs centralized, holdout test dataset

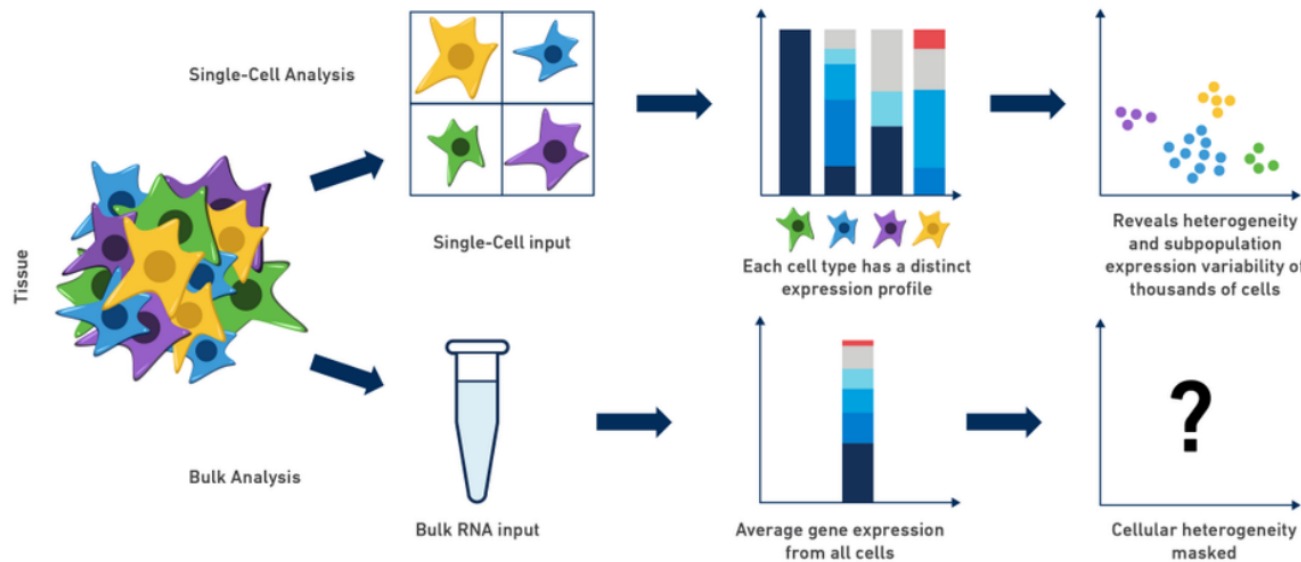


B Federated vs centralized, external test dataset



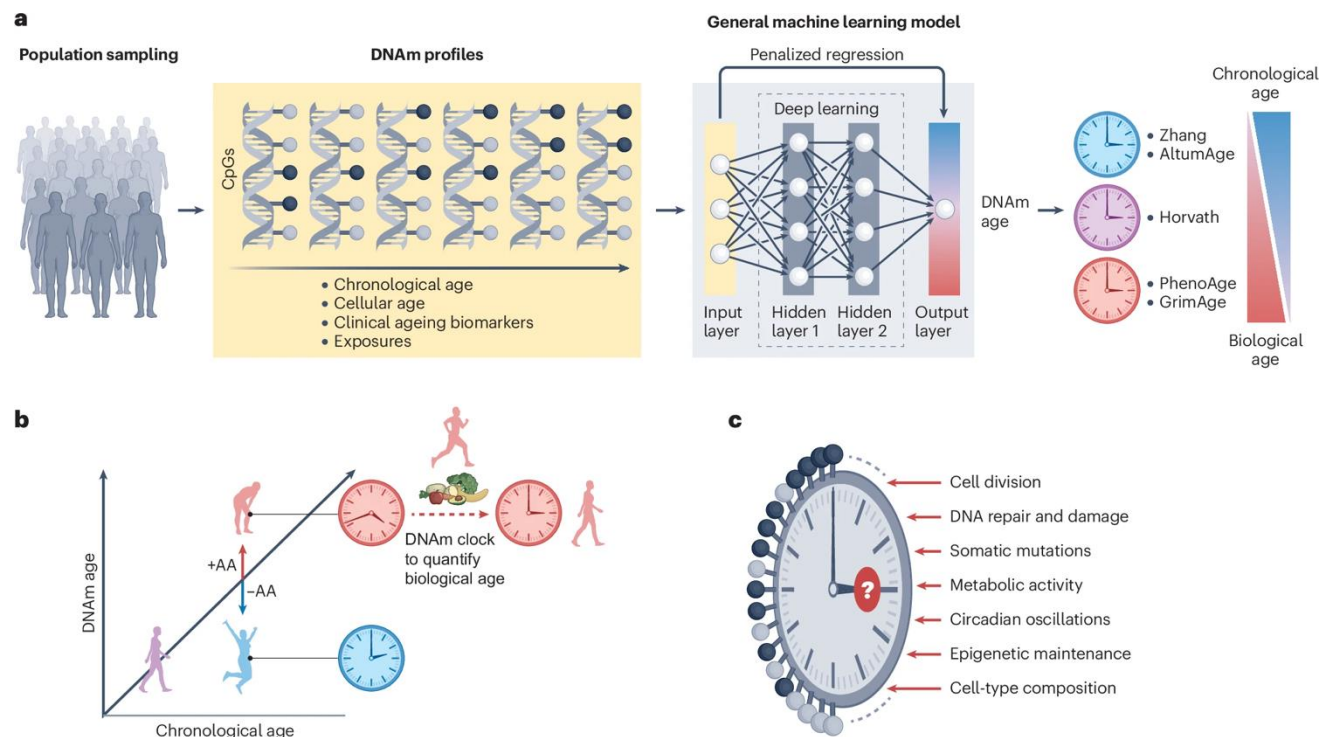
- Holdout (Data belonging to the participant facilities i.e. involved in the training)
- External (Data from an external hospital, not involved in the training)

Single – cell RNA-sequencing (scRNA-seq)



- Captures transcriptomes at the resolution of individual cells, revealing cell-to-cell variability hidden in bulk RNA-seq.
- Identifies **distinct cell types and subtypes** within a mixed population.
- Detects **cell states** (e.g., activation, stress, differentiation) within the same cell type.
- Quantifies **intratumoral heterogeneity** in cancer research, informing treatment resistance studies.

Transcriptomics ageing clocks

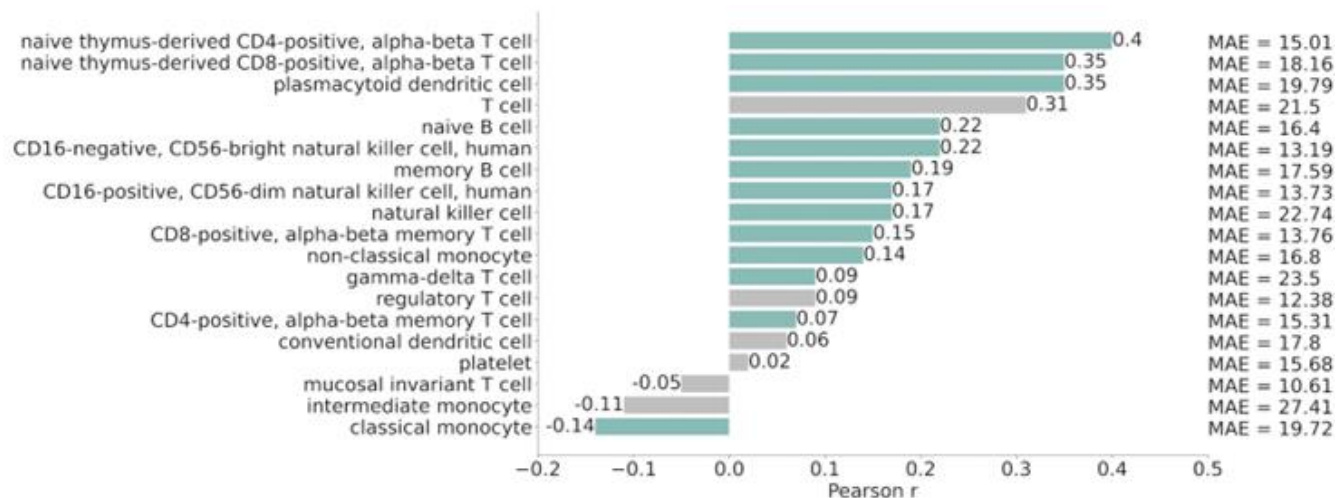


Biomarkers that estimate biological age rather than chronological age.

Typically based on DNA methylation patterns at specific CpG sites but other omics exists (proteomics transcriptomics).

Next-generation clocks: incorporate health, lifestyle, and mortality predictors (e.g., PhenoAge, GrimAge).

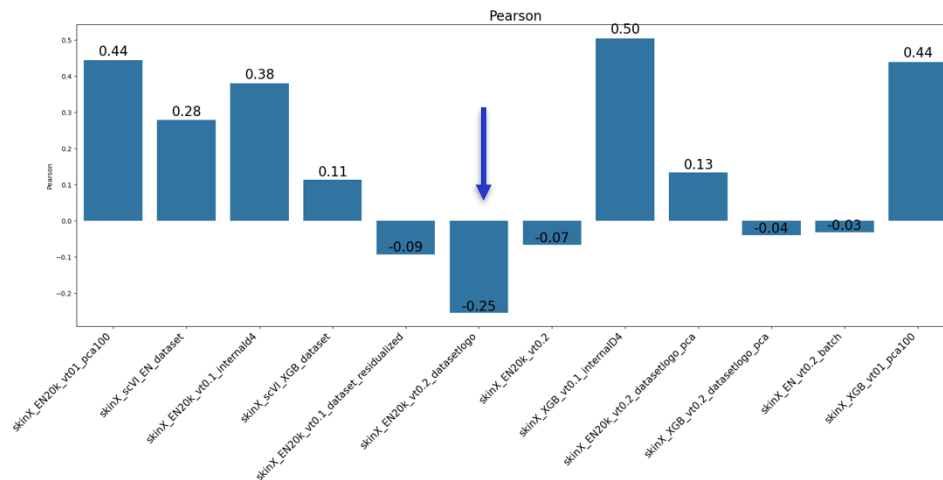
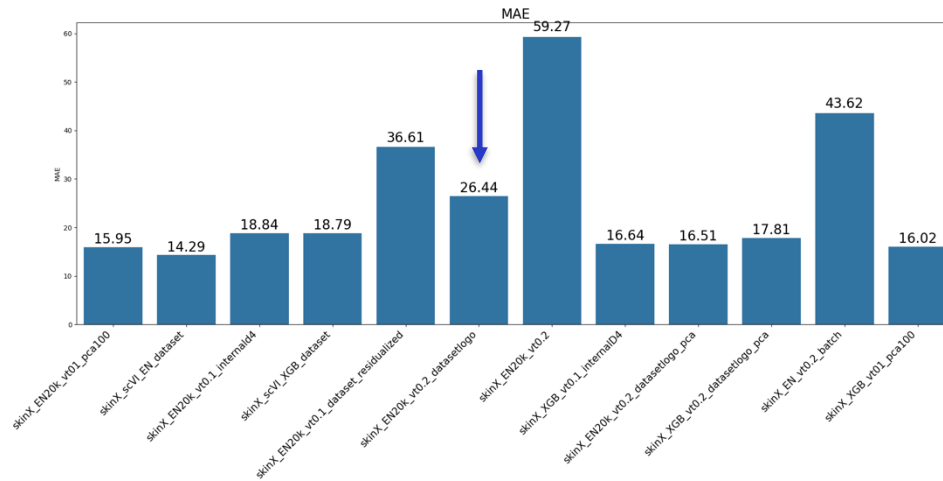
Transcriptomics ageing clocks – state of the art



Model trained on 1 million cells from blood

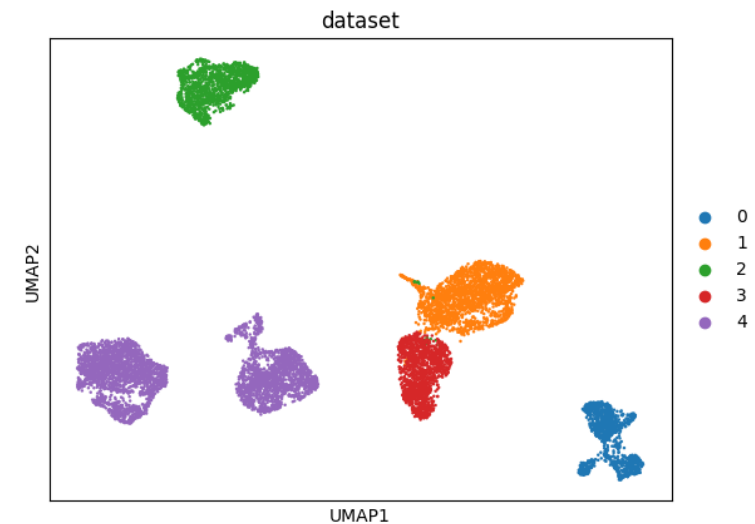
For many cell populations the model didn't give precise predictions

Our results – even worse

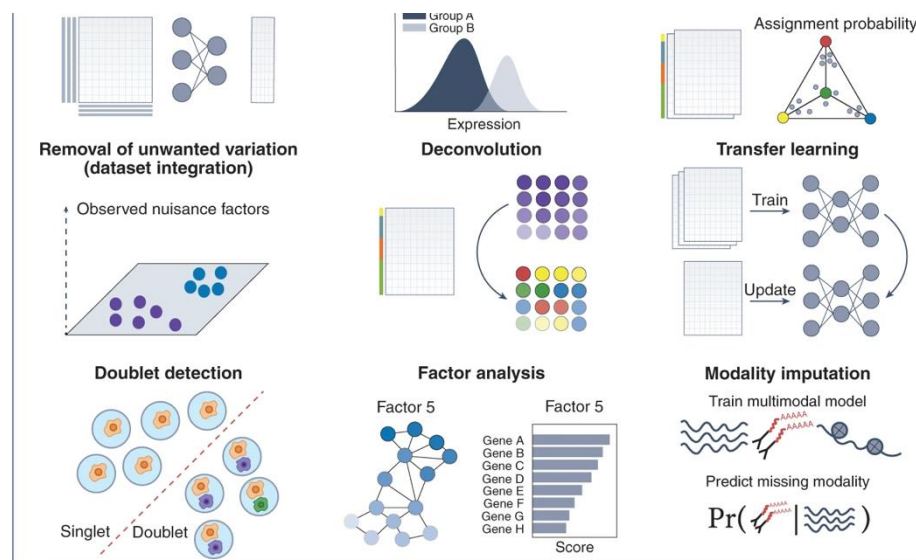
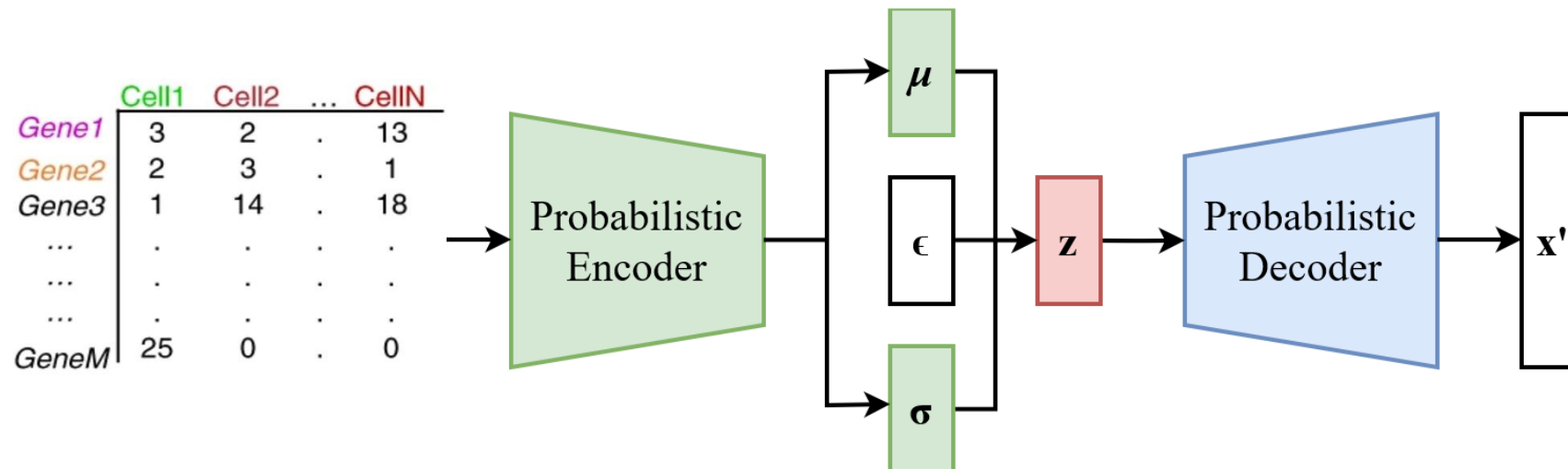


We tried to mimick the results using another cell population for which we have less data

Thus we thought of integrating different datasets.



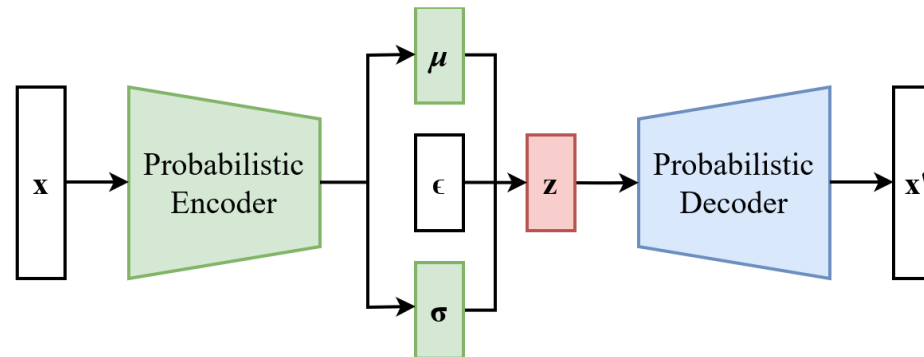
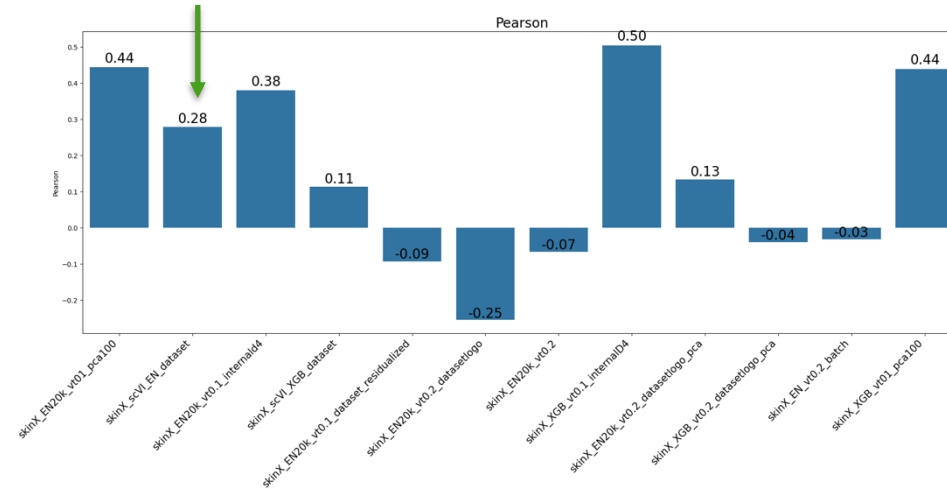
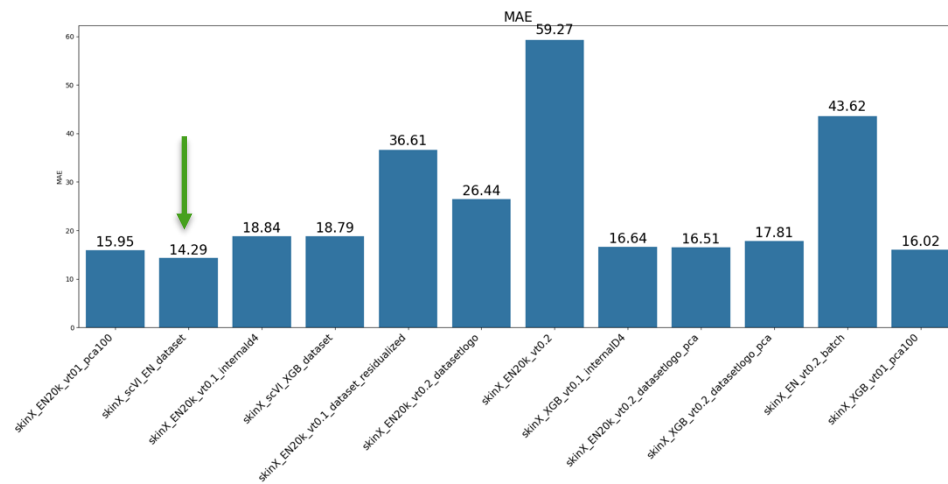
scVI toolkit



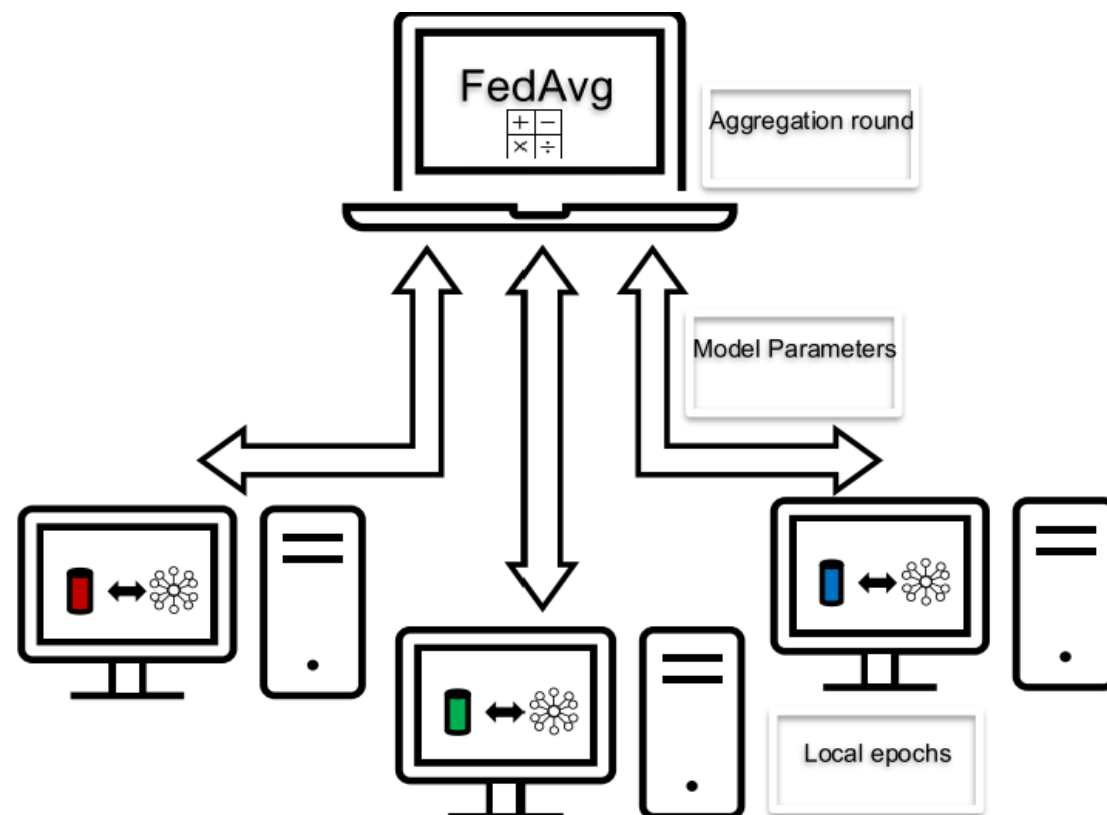
Downstream tasks:

- Batch effect Mitigation
- Reference Mapping
- DEG

Transcriptomics ageing clocks – Latent embeddings



Federated – scVIC



Federated scVI

- **Objective:** learn a clean latent for cells.
- **Data:** count matrices + batch label (batch_key).
- **Training :** ELBO = Expected NB reconstruction – KL divergence regularization
- **Outputs:** batch-corrected latent for UMAP/clustering

IN SUMMARY

- Nowadays, hospitals generate and store massive amounts of data, creating major opportunities for integrating AI into biology and medicine.
- However, these data are highly sensitive and often costly to generate, raising significant privacy and data-sharing concerns.
- Federated learning offers a potential solution to mitigate privacy risks by enabling collaborative model training without direct data sharing, though it may be less effective in addressing cost-related challenges.
- Its effectiveness is proven, federated approaches can perform comparably to centralized learning and outperform traditional meta-analysis methods.
- Further progress is needed not only on the technological front, but also in regulatory and governance frameworks, to support broader adoption of federated learning